

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 47

WORLD CLIMATIC TYPES
**THE TROPICAL MONSOON &
MARINE CLIMATE**

THE TROPICAL MONSOON AND TROPICAL MARINE CLIMATES

GS-1 2017 QUESTION

Qn

What characteristics can be assigned to monsoon climate that succeeds in feeding more than 50 percent of the world population residing in Monsoon Asia? (250 words)

WORLD CLIMATIC TYPES

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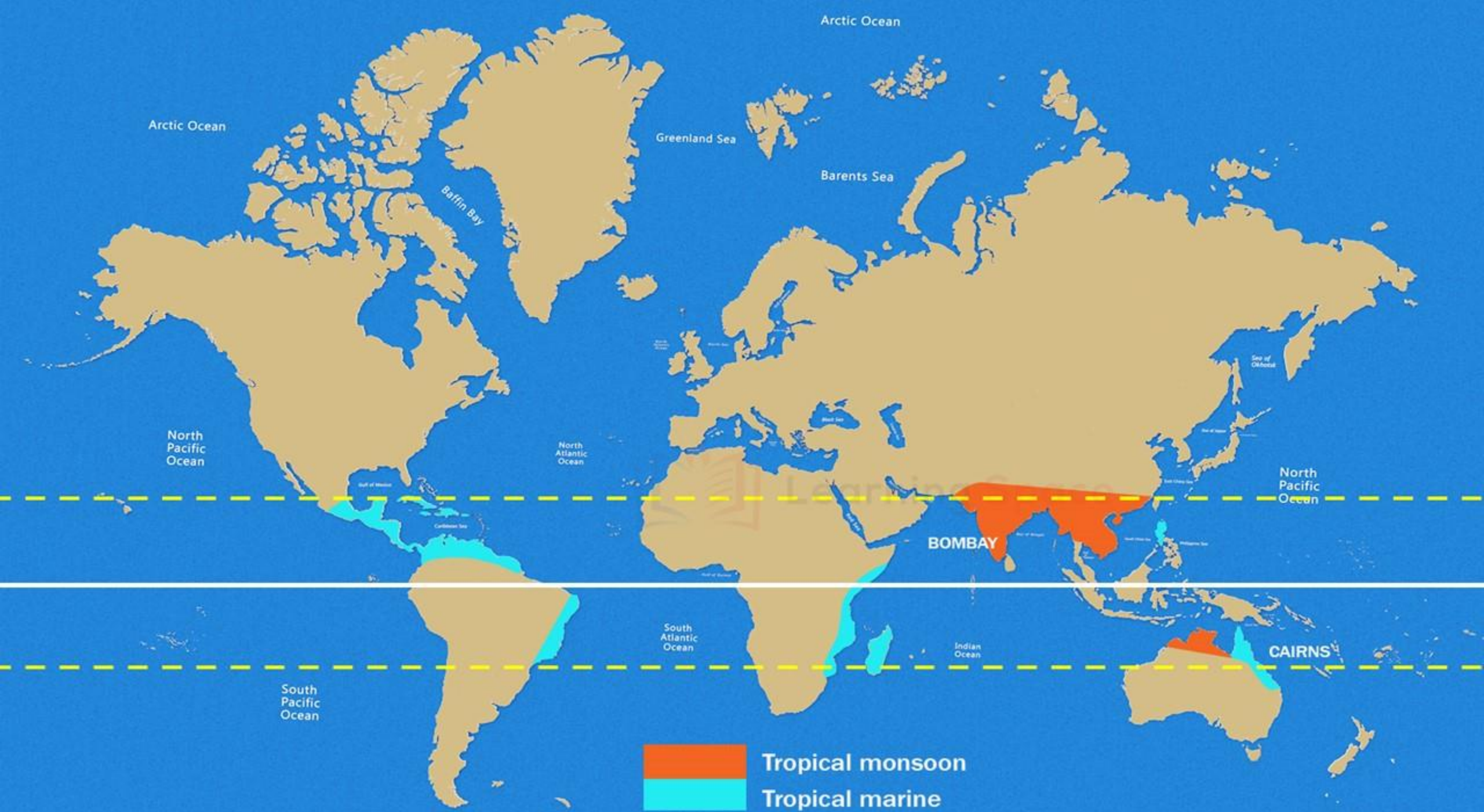
THE TROPICAL MONSOON AND TROPICAL MARINE CLIMATES

DISTRIBUTION

1 Tropical Monsoon Climate are best developed in the Indian sub-continent, Burma, Thailand, Laos, Cambodia, parts of Vietnam and south China and northern Australia.

2 Outside this zone, the climate is modified by the influence of the on-shore Trade Winds all the year round and has a more evenly distributed rainfall.

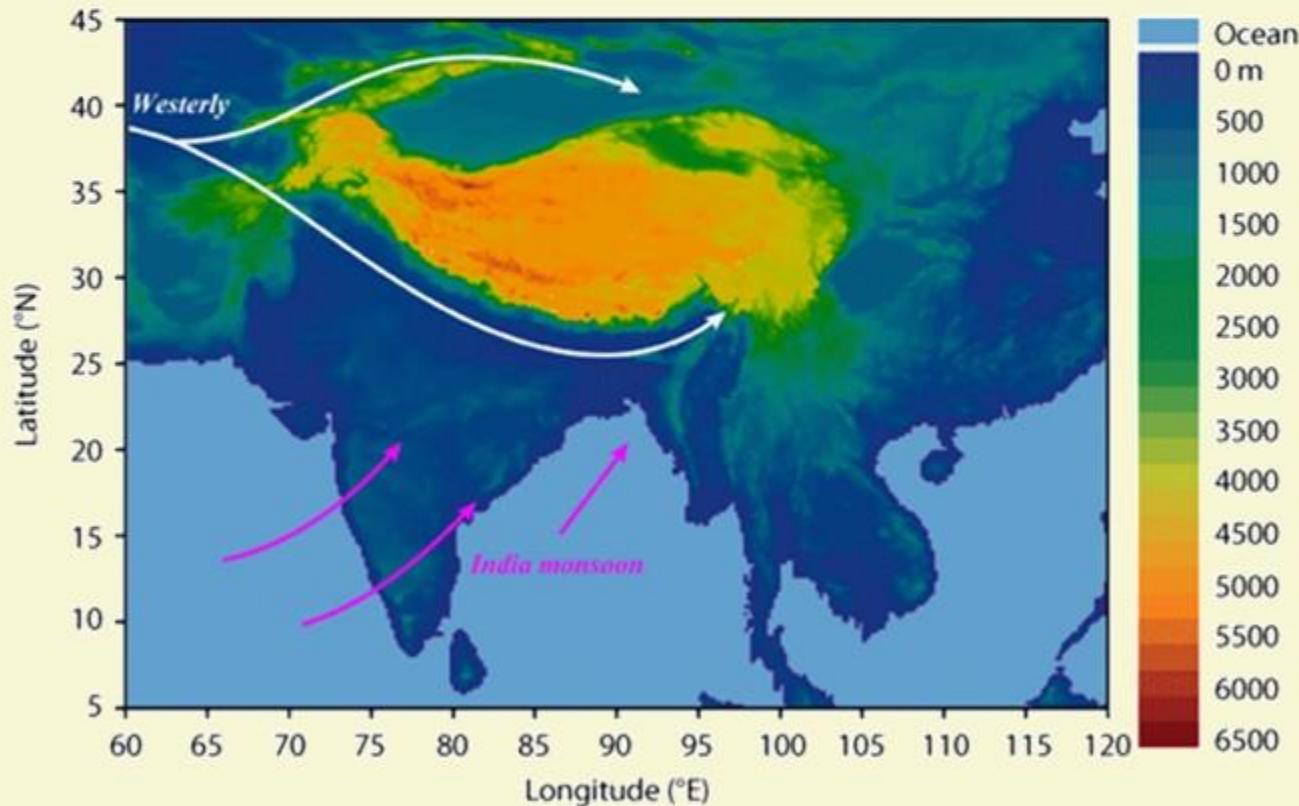
3 Such a climate, better termed the Tropical Marine Climate, is experienced in Central America, West Indies, north-eastern Australia, the Philippines, parts of East Africa, Madagascar, the Guinea Coast and eastern Brazil.



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THE TROPICAL MONSOON AND TROPICAL MARINE CLIMATES

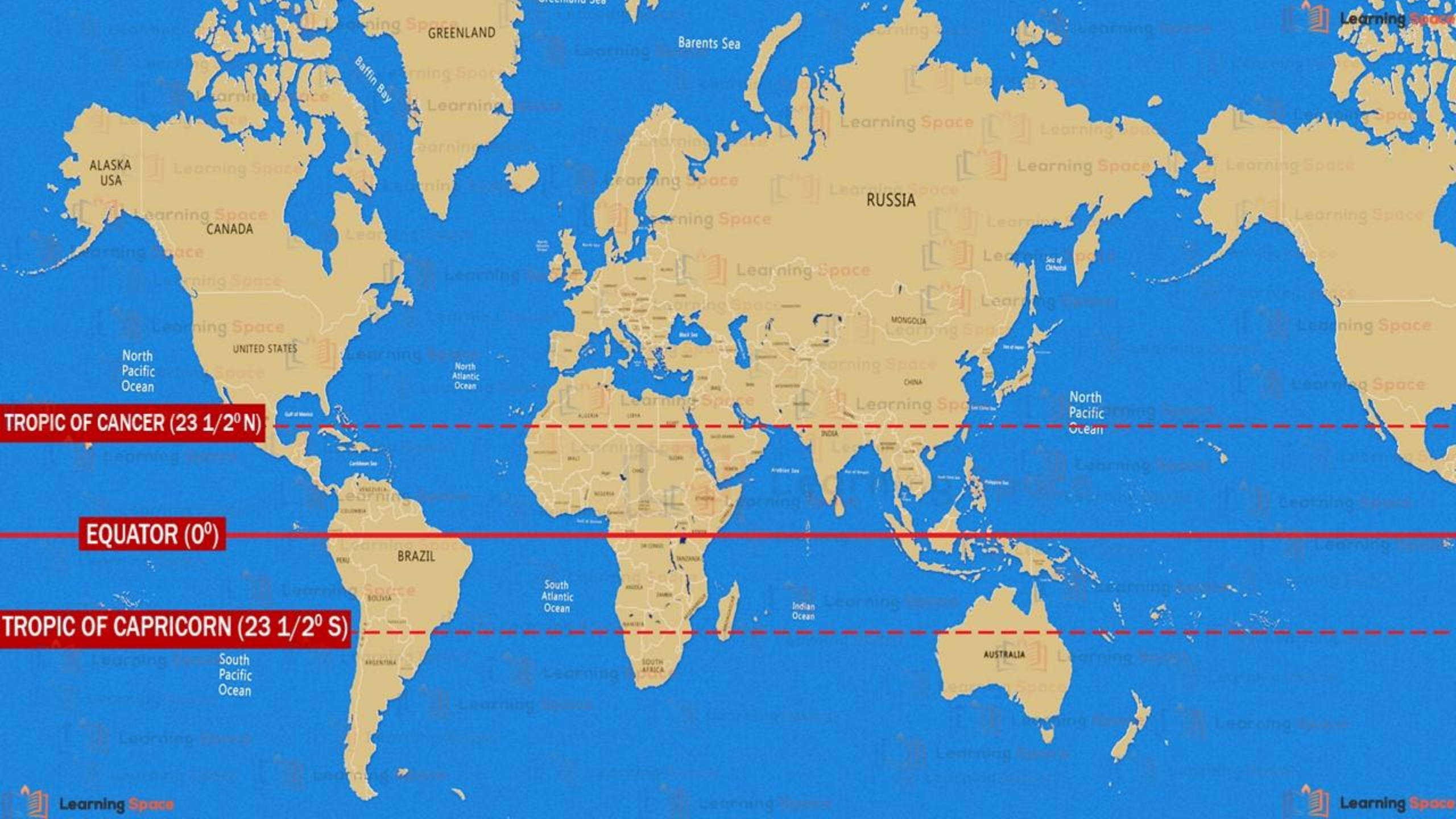
CLIMATIC CONDITIONS IN TROPICAL MONSOON LANDS



1 The basic cause of monsoon climates is the **difference in the rate of heating and cooling of land and sea.**

2 In the summer, when the sun is overhead at the Tropic of Cancer, the great land masses of the northern hemisphere are heated.

3 **Central Asia**, backed by the lofty Himalayan ranges, is more than 15°F hotter than its normal temperature and a **region of intense low pressure is set up.**



TROPIC OF CANCER ($23\frac{1}{2}^{\circ}\text{N}$)

EQUATOR (0°)

TROPIC OF CAPRICORN ($23\frac{1}{2}^{\circ}\text{S}$)

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THE TROPICAL MONSOON AND TROPICAL MARINE CLIMATES

CLIMATIC CONDITIONS IN TROPICAL MONSOON LANDS

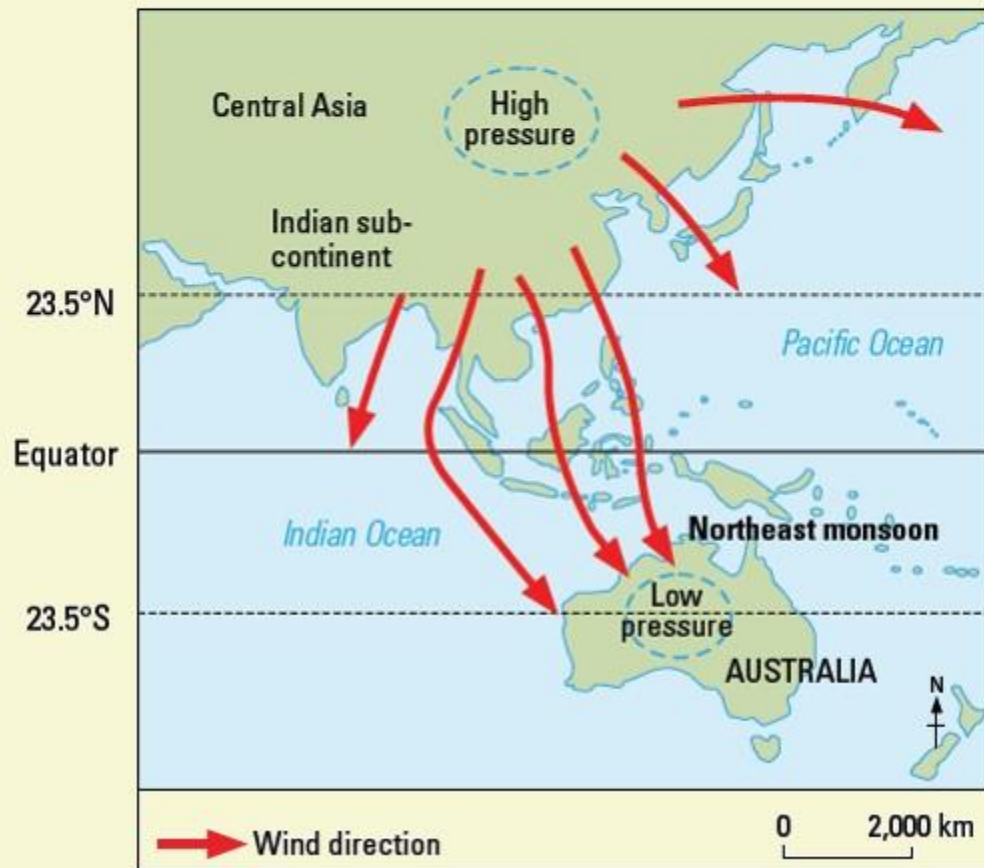


- 4 The seas which warm up much slower, remain comparatively cool.
- 5 At the same time, the southern hemisphere experience winter, and a region of high pressure is set up in the continental interior of Australia.
- 6 Wind blow outwards as the South-East Monsoon, to Java and after crossing the equator are drawn towards the continental low pressure area reaching the Indian sub-continent as the **South-West Monsoon**.

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THE TROPICAL MONSOON AND TROPICAL MARINE CLIMATES

CLIMATIC CONDITIONS IN TROPICAL MONSOON LANDS



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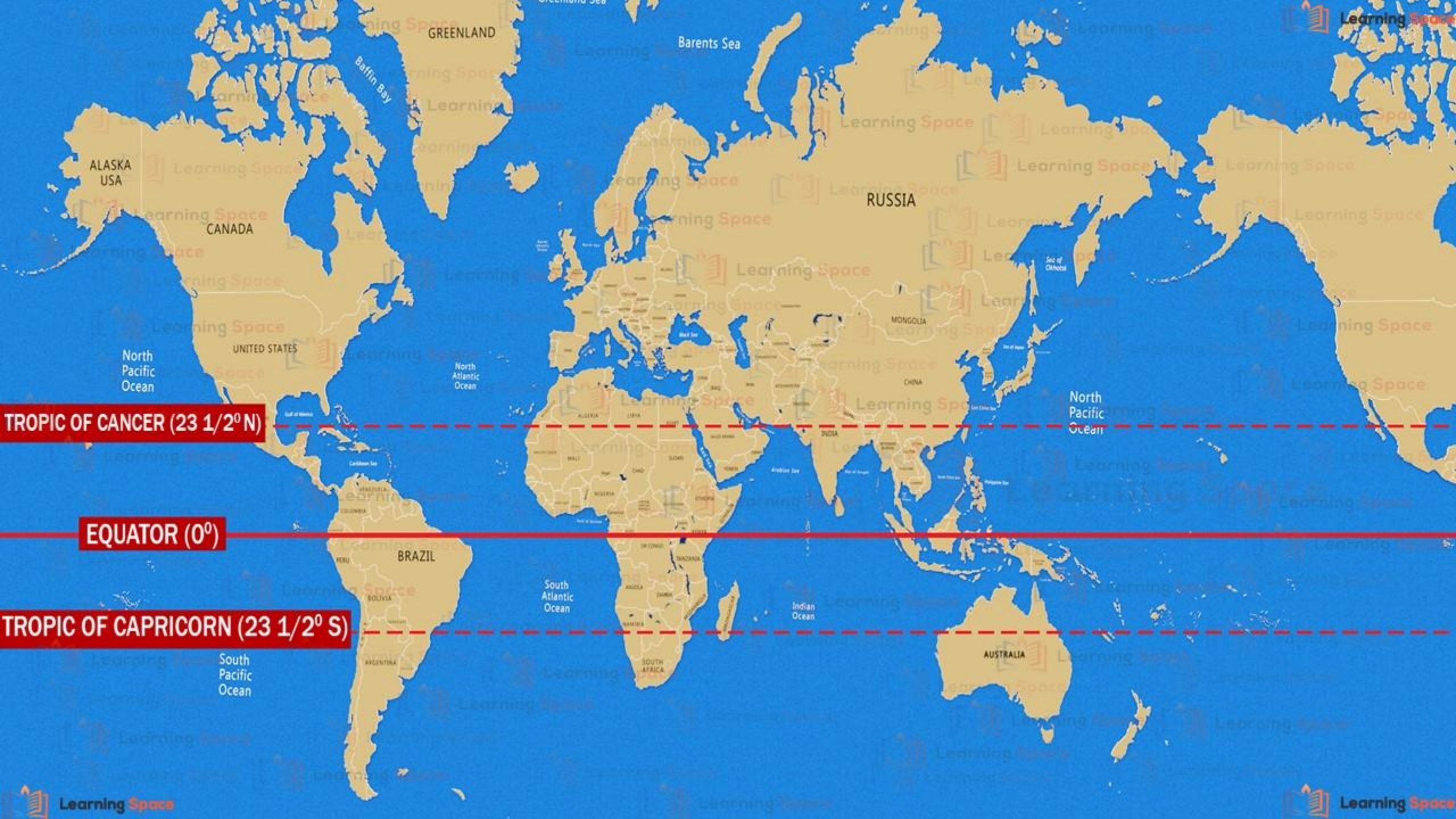
In the winter, conditions are reversed. The Sun is overhead at the Tropic of Capricorn, **Central Asia** is extremely cold, resulting in rapid cooling of the land.

8

A region of high pressure is created with out blowing winds the **North-East Monsoon**.

9

On crossing the equator, the winds are attracted to the low pressure center in Australia and arrive in northern Australia as the **North-West Monsoon**.



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THE TROPICAL MONSOON AND TROPICAL MARINE CLIMATES

THE SEASONS OF TROPICAL MONSOON CLIMATE



In regions like the Indian sub-continent which have a true Tropical Monsoon Climate, the following seasons are generally found:

1. THE COOL DRY SEASON (October to February)
2. THE HOT DRY SEASON (March To Mid-June)
3. THE RAINY SEASON (Mid-June to September)

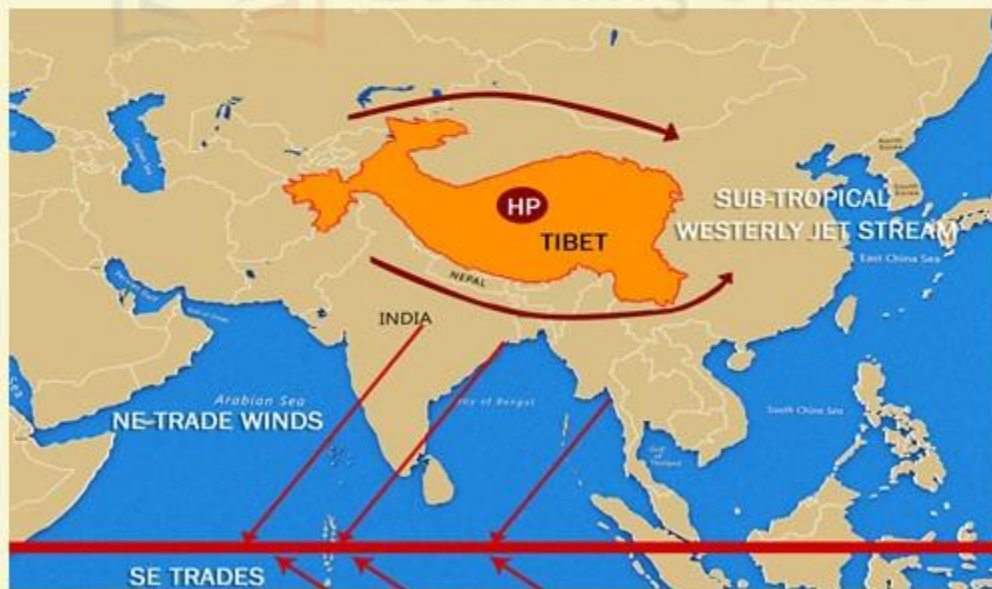
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THE TROPICAL MONSOON AND TROPICAL MARINE CLIMATES

THE SEASONS OF TROPICAL MONSOON CLIMATE

THE COOL, DRY SEASON (OCTOBER TO FEBRUARY)

- 1 Out blowing dry winds, the North-East Monsoon, bring little or no rain to the Indian sub-continent.



North-East Monsoon

- 2 However, a small amount of rain falls in Punjab from cyclonic sources and this is vital for the survival of winter cereals.
- 3 Where the North-East Monsoon blows over the Bay of Bengal it acquires moisture and thus brings rain to the south-eastern tip of the peninsula at this time of the year.
- 4 For instance, in Madras 50 inches of rain falls during October and November, accounting for half its annual rainfall.

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THE TROPICAL MONSOON AND TROPICAL MARINE CLIMATES

THE SEASONS OF TROPICAL MONSOON CLIMATE

THE HOT DRY SEASON (MARCH TO MID-JUNE)



1 The temperature rises sharply with the sun's northward shift to the tropic of Cancer.

2 Day temperatures of 95°F are usual in Central India

3 Coastal districts are a little relieved by sea breezes. There is practically no rain anywhere.

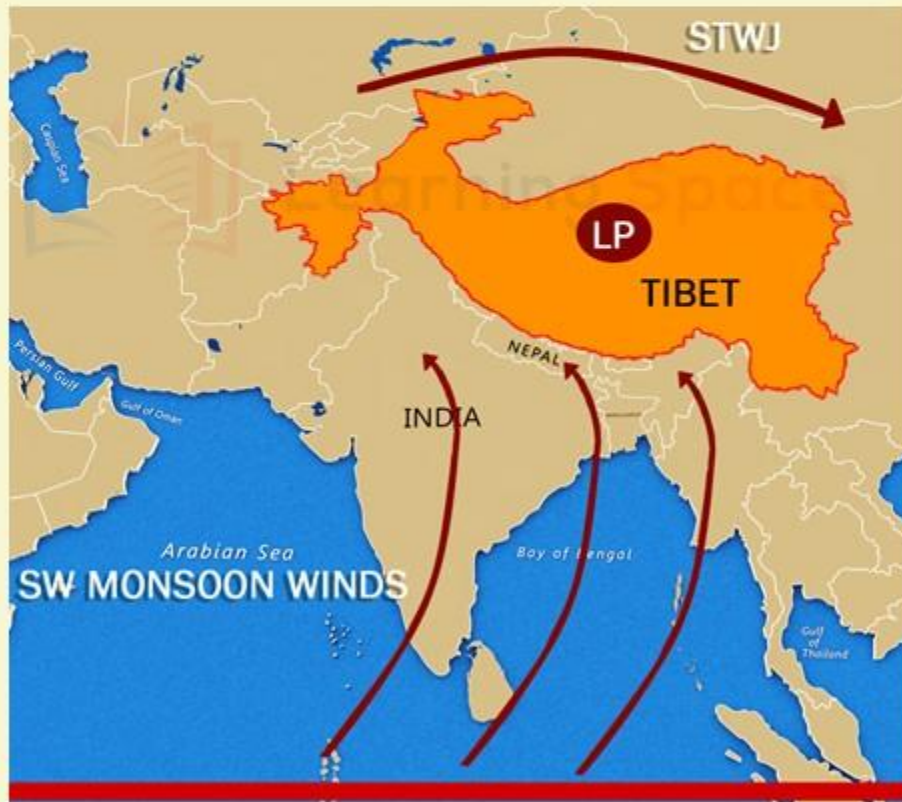
4 Dust storms are also frequent.

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THE TROPICAL MONSOON AND TROPICAL MARINE CLIMATES

THE SEASONS OF TROPICAL MONSOON CLIMATE

THE RAINY SEASON (MID-JUNE TO SEPTEMBER)



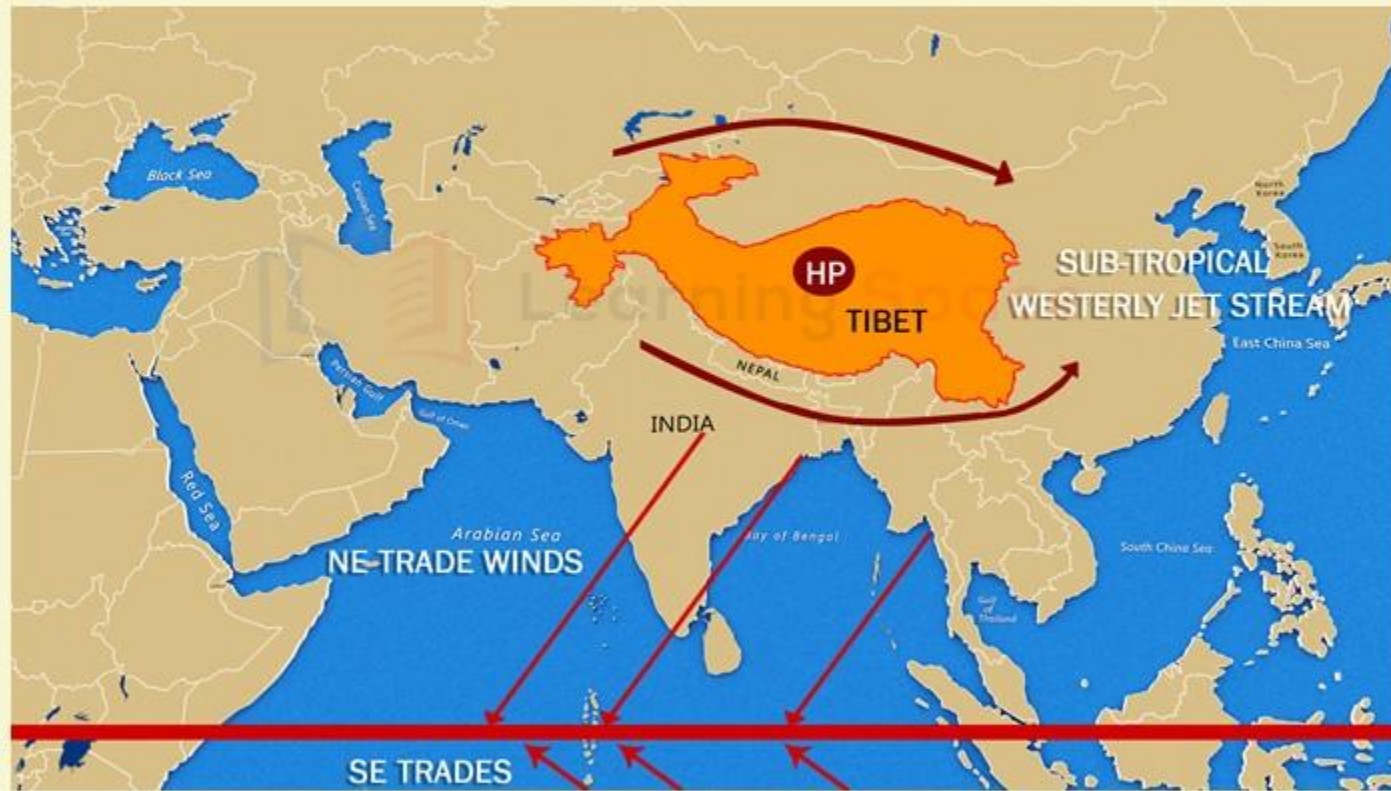
South-West Monsoon

- 1 With the “burst” of the South-West Monsoon in mid-June, torrential downpours sweep across the country to the delight of everybody.
- 2 Almost all the rain for the year falls within this rainy season.
- 3 This pattern of concentrated heavy rainfall in summer is a characteristic feature of the Tropical Monsoon Climate.

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THE TROPICAL MONSOON AND TROPICAL MARINE CLIMATES

THE RETREATING MONSOON



North-East Monsoon in the Retreating Monsoon Season

- 1 The amount and frequency of rain decreases towards the end of the rainy season.
- 2 It retreats gradually southwards after mid-September until it leaves the continent altogether.
- 3 The Punjab plains are the first to see the withdrawal of the monsoon.
- 4 The skies are clear again and the cool, dry season returns in October, with the out blowing North-East Monsoon.

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THE TROPICAL MONSOON AND TROPICAL MARINE CLIMATES

THE RETREATING MONSOON



Floods of Kerala

- 5 The role of monsoons in India is vital in its economy. A late monsoon or one that ends far too early will condemn large stretches of agricultural land to drought.
- 6 There will be widespread famine from crop failure and thousand will perish.
- 7 When there is too much water from the rainy monsoons, severe floods occur, destroying both crops and lives and disrupting communications.
- 8 In no part of the world has the climate affected Man's way of life so profoundly as in the monsoon lands.

THE TROPICAL MONSOON AND TROPICAL MARINE CLIMATES

GS-1 2017 QUESTION

Qn

What characteristics can be assigned to monsoon climate that succeeds in feeding more than 50 percent of the world population residing in Monsoon Asia? (250 words)

DOS AND DON'TS

- 1** Don't mention about the mechanism of Monsoon Climate. The examiner wants us to come up with the characteristics of monsoon climate which has a profound impact on the agricultural cycle of the region.
- 2** Write about the defining features of Monsoon Climate- the seasons along with their months and their associated character in crisp points.
- 3** Write about the vegetation of the region- deciduous character, varies from forests to scrublands. Mention some important species.
- 4** Write about diversity in temperature, precipitation in every region and in spite of all this, how agriculture is a thriving economy

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